

develop a wide variety of brain implants. At first, these would undoubtedly serve to improve treatment of diseases with root causes in the brain, such as Parkinson's disease and epilepsy. Implants might also be developed to replace parts of the brain damaged by stroke or injury, or to fine-tune imbalances that cause severe depression and other mental illnesses. Eventually, implants might be used to enhance normal brain function—for example, to boost memory and learning. Of course, at each step there is likely to be a lot of soul searching and debate about the ethics of these interventions and about what it means to be human.

Scientists predict four distinct revolutions in medicine, and envision that nanotechnology could play a pivotal role in driving any or all of them. Nanotechnology could be used to make medicine more predictive, preemptive, personalized and participatory (regenerative).

1. Predictive medicine- In this vision nanotechnology would help doctors predict the major diseases that an individual is likely to develop. The goal would be to routinely and cheaply analyze several hundred substances in a patient's blood and estimate disease risks with a relatively high degree of accuracy. This panel of tests would provide a window on a person's overall state of health.

As mentioned earlier, several research groups working on developing a “lab-on-a-chip” device—using nanotechnology—to perform a comprehensive analysis of a drop of blood. The blood analysis would alert the doctor to early precursors of disease that reflect both genetic predispositions and environmental factors, such as diet, exercise, stress and exposure to air pollution. Predictive medicine sounds incredibly promising, but developing such a comprehensive test might prove to be exceedingly challenging.

- ▶ First, researchers would have to identify the 200 or so most important disease biomarkers.
- ▶ Next, although scientists would not have to understand every last detail about each biomarker and its role in disease, they would have to know how the levels of each biomarker vary normally and in various stages of disease.
- ▶ Patterns will have to be identified by trolling through huge amounts of data collected from a road population sample. And the computations might require fancy mathematical tools that have not yet been invented.

The complexities don't end here. For



Nanomedicine